The Derailing of the MBTA: A Data-Driven Analysis

1. **Problem Statement:** The Massachusetts Bay Transportation Authority (MBTA) is critical for many people as they want to get around the Greater Boston area. For this final project, I address the following question to evaluate the system's performance:

   Is the MBTA providing reliable, predictable, and satisfactory service across lines and time periods?

2. **Motivation**: My motivation for this analysis comes from my personal experience as a resident of a Boston suburb. I often use the MBTA to avoid traffic when traveling into the city. However, I have encountered an abundance of difficulties in trying to navigate in and out of Boston efficiently while using the MBTA.

   These personal frustrations prompted me to move beyond my own experiences and study how reliable the MBTA is as a whole. By analyzing the data, I aim to quantify the disruptions and satisfaction levels that riders experience daily.

3. **Data Source 3.1 Dataset Overview**: To do this analysis, I used four datasets from MassDOT public records through their open data portal:

Service Alerts: Information regarding system notifications and interruptions.

Expected vs. Actual Arrival Times: Data allowing for the calculation of delays and prediction accuracy.

Rider Satisfaction Survey: Direct feedback from passengers regarding their experience.

Work Orders: Operational data detailing maintenance efforts, hours worked, and costs.

3. **3.2 Deeper Details and Outliers:** Through analysis using Python, patterns and outliers were identified from the Work Orders data.

Operational Costs: The total cost of jobs in the dataset amounted to over $3 million with an average job cost of $303.42.

Labor: The average crew hours worked per job is 7.57 hours.

Outliers: Box plots for crew hours and material costs revealed significant outliers. While the median crew hours are relatively low (around 7 hours), there are times where crews worked up to 37 hours. Similarly, while the median material cost is lower, there are outliers reaching nearly $2,000 for a single job.

4. Story

4.a. Introduction: The Rider Perspective: The narrative begins not with the trains themselves, but with the people who ride them. The anonymous satisfaction survey sent to people after riding the MBTA reveals a striking detail in public opinion. When asked to rate their most recent trip, the majority of people were overall pleased. However, when asked if the MBTA is reliable, the responses are much more negative (Figure 1). This sets the stage: riders generally have decent individual trips, but the total effect of inconsistency has damaged their trust in the system.

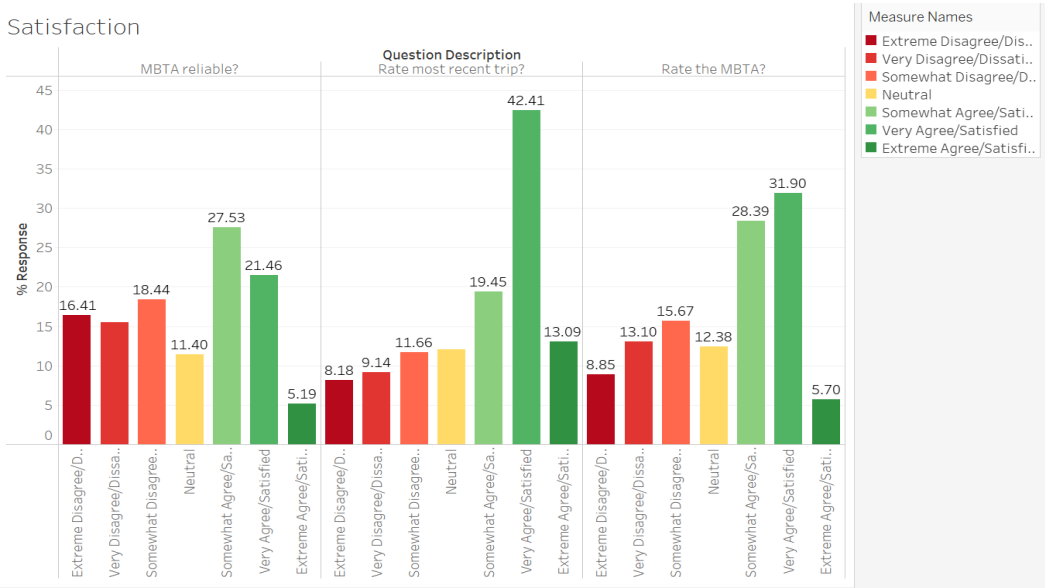

Figure 1

4.b. Core Evidence: To understand this gap in trust, I analyzed the volume and severity of service alerts (Figure 2).

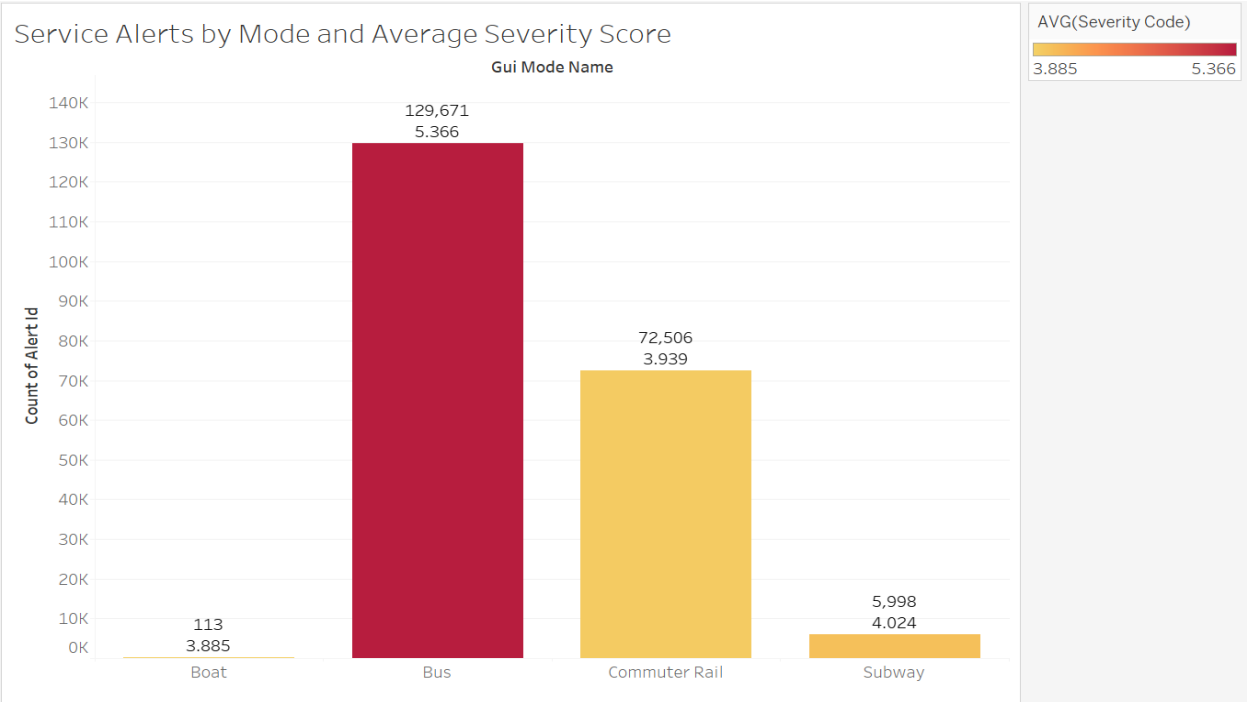

Figure 2

Differences in Mode: The Bus system is the primary source of system alerts, with 129,671 unique alerts and a high average severity score of 5.366. By comparison, the Subway has far fewer alerts (5,998) but maintains a decently high severity score (4.024).

Bus Issues: For buses, 91.75% of alerts are classified as "Delay." The tree map of causes shows that this is mostly an external problem as "Traffic" is the dominant cause (Figures 3 & 4).

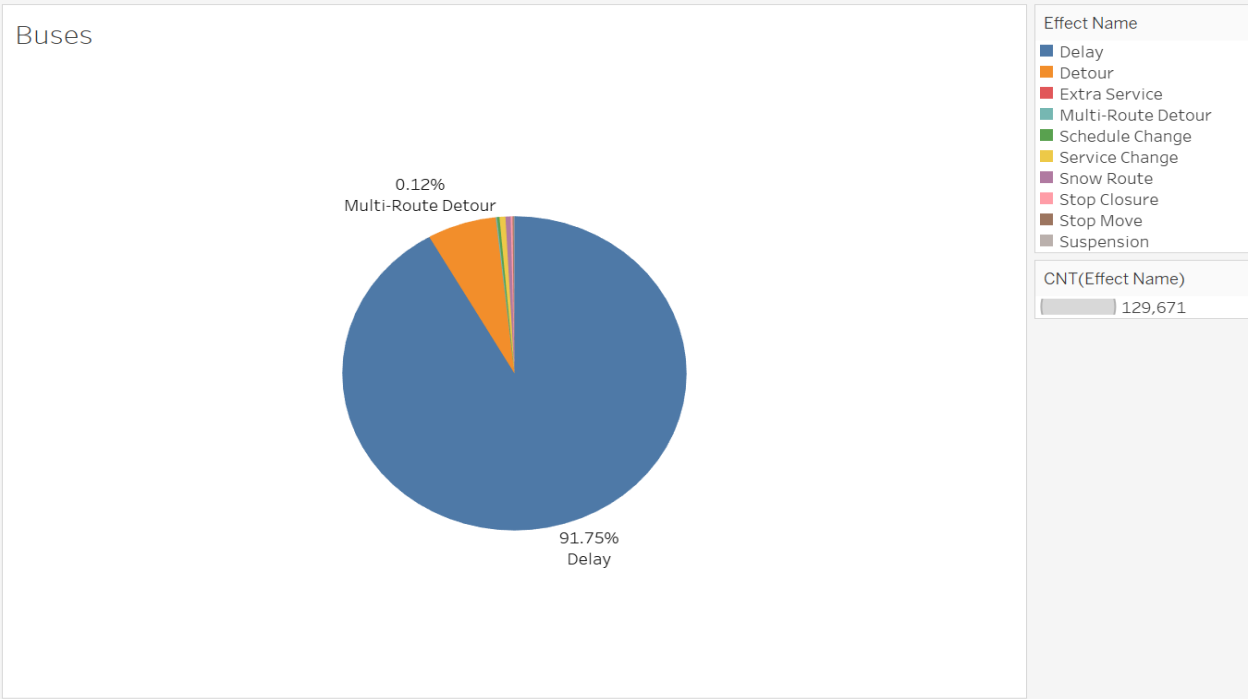

Figure 3

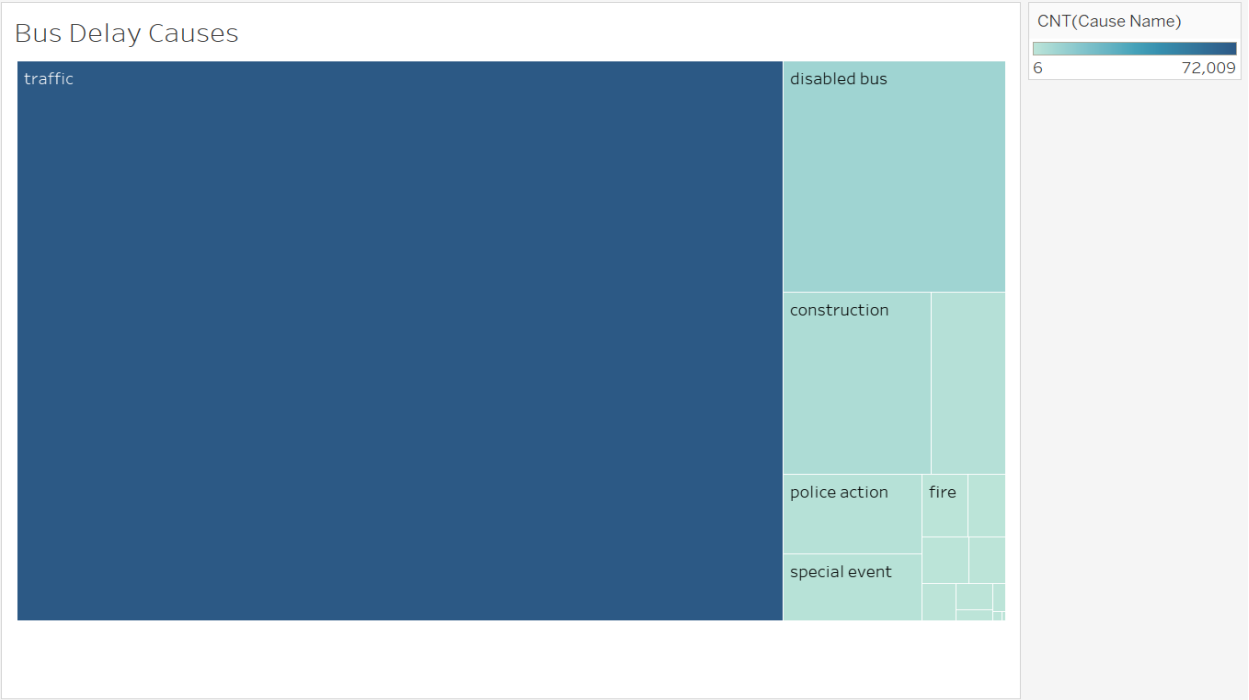

Figure 4

Rail Issues: The rail system tells a different story. While delays are still the top effect (79.21%), "Track Change" appears as a significant issue. More importantly, the delay causes within the rail

system are internal. "Disabled Train" and "Signal Problem" are the largest in the tree map, showing infrastructure failure (Figures 4 & 5).

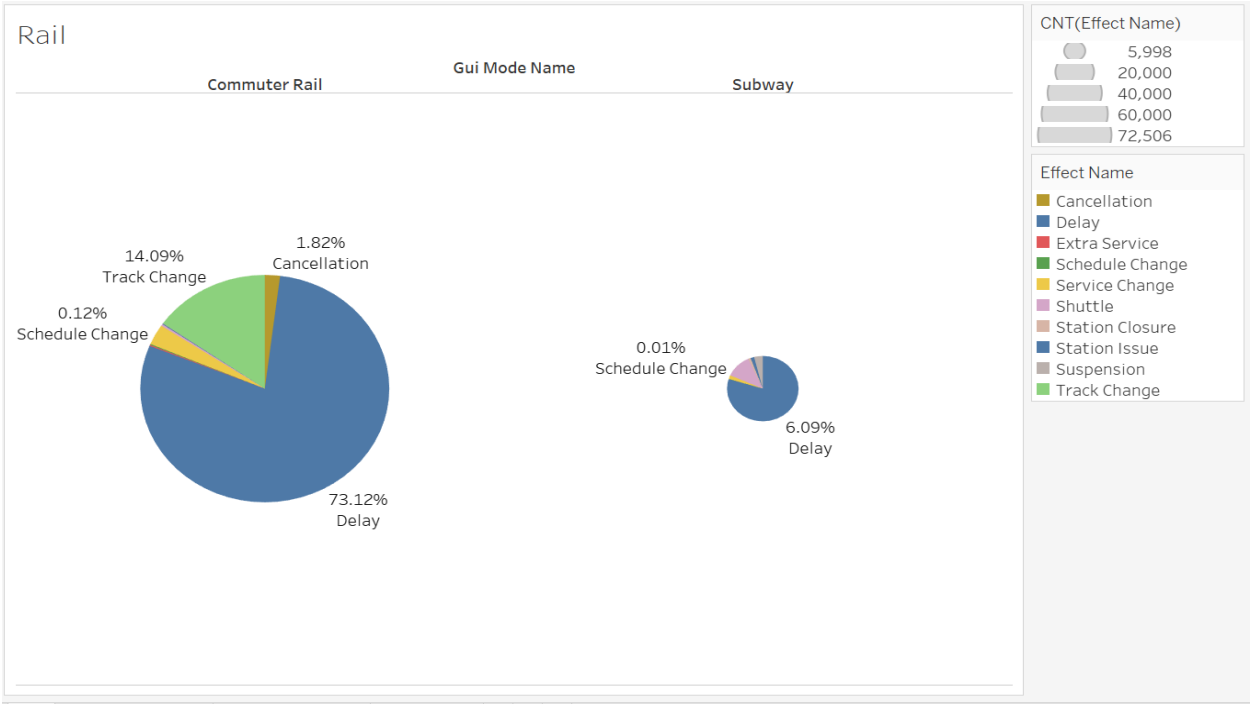

Figure 4

Figure 5

Predictability: The trend of predictions over time shows a consistent gap between the total number of predictions made and those that were accurate, suggesting that the system could use improvements in this area. An incorrect prediction could be the difference between someone making it to work on time or not (Figures 6 & 7).

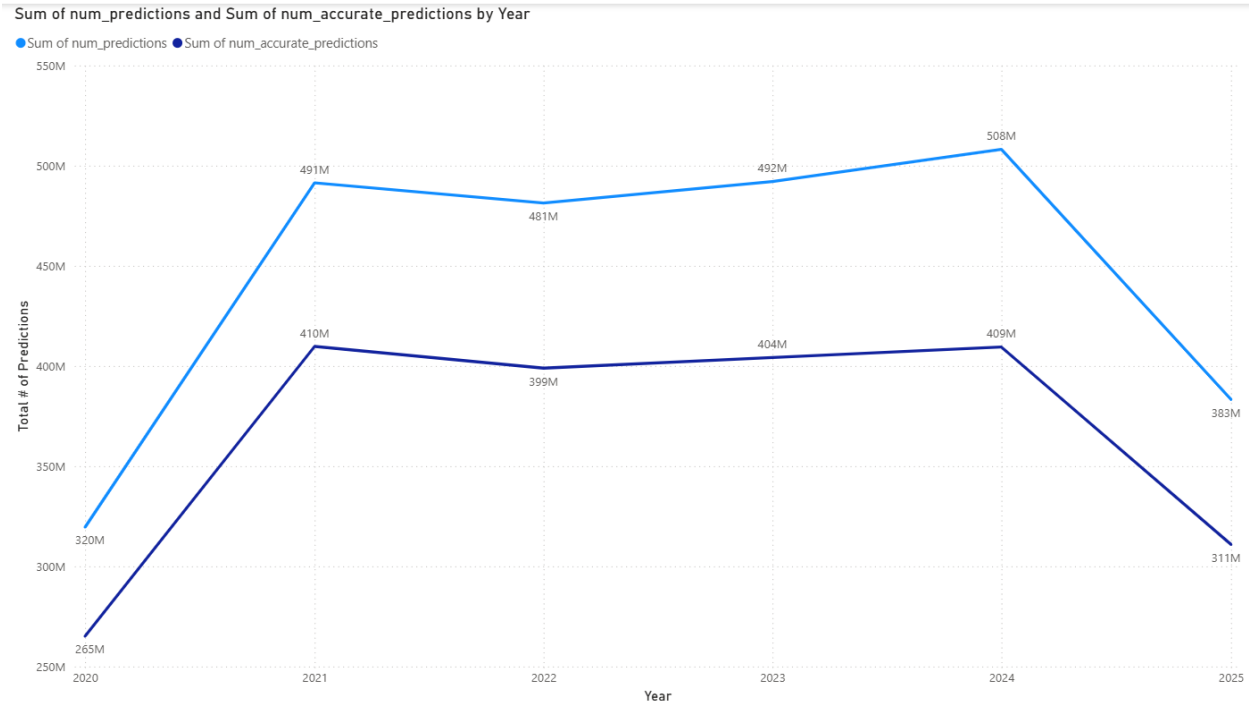

Figure 6

Figure 7

Line & Stop Performance: Breaking this down by line, the Blue Line surprisingly shows the highest average delay, but the others all followed closely behind (Figure 8).

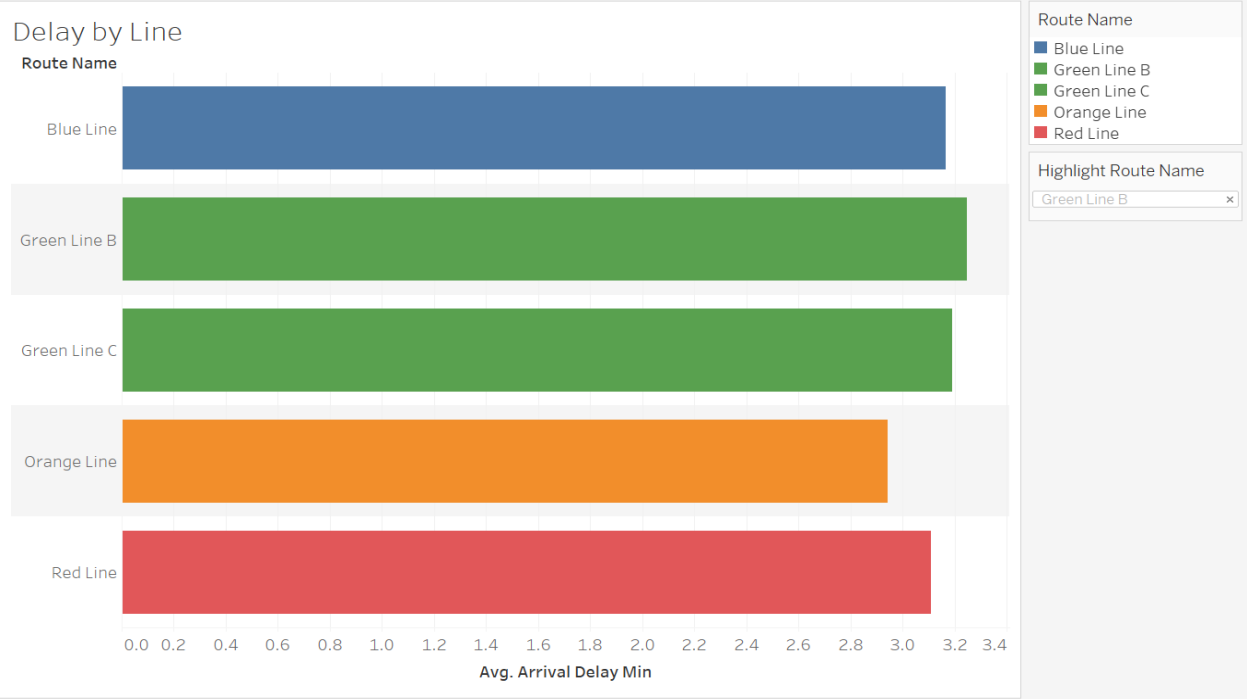

Figure 8

Major stops like Harvard and Porter Square experience higher average delays. This is critical because ridership data shows Park Street as having the highest passenger volume. If delays happen in these high-volume hubs, the impact on riders is extreme (Figures 9 & 10).

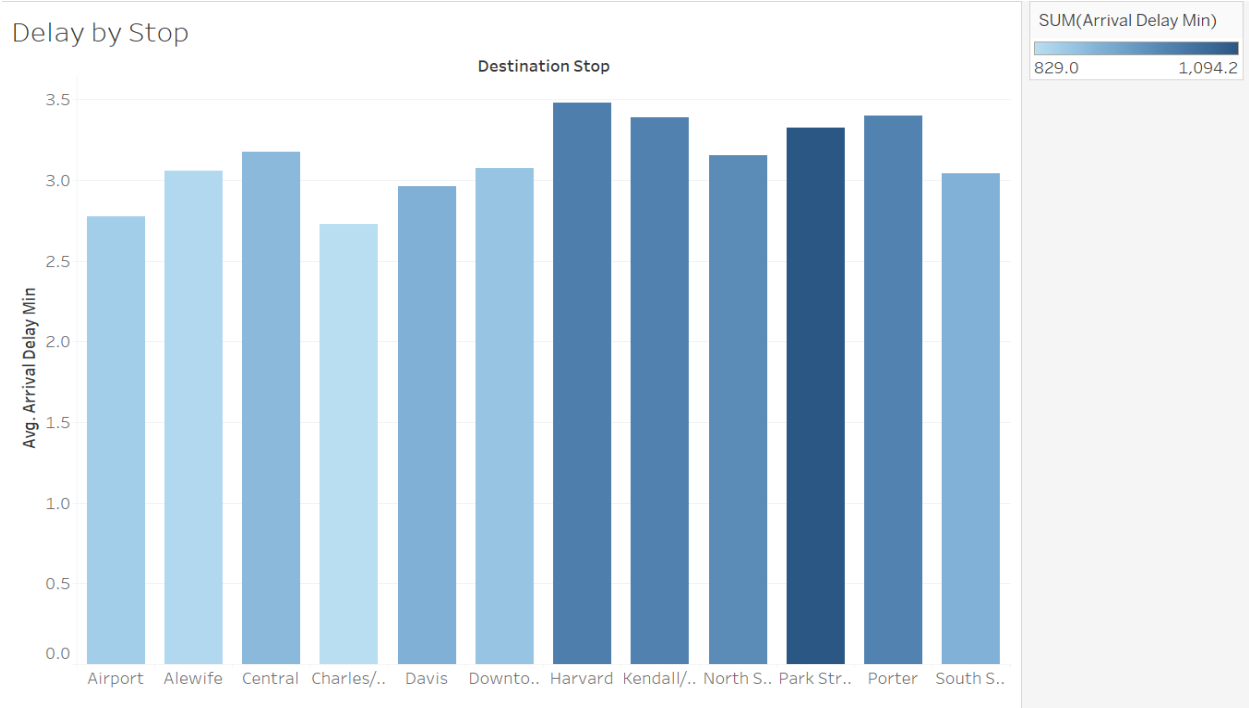

Figure 9

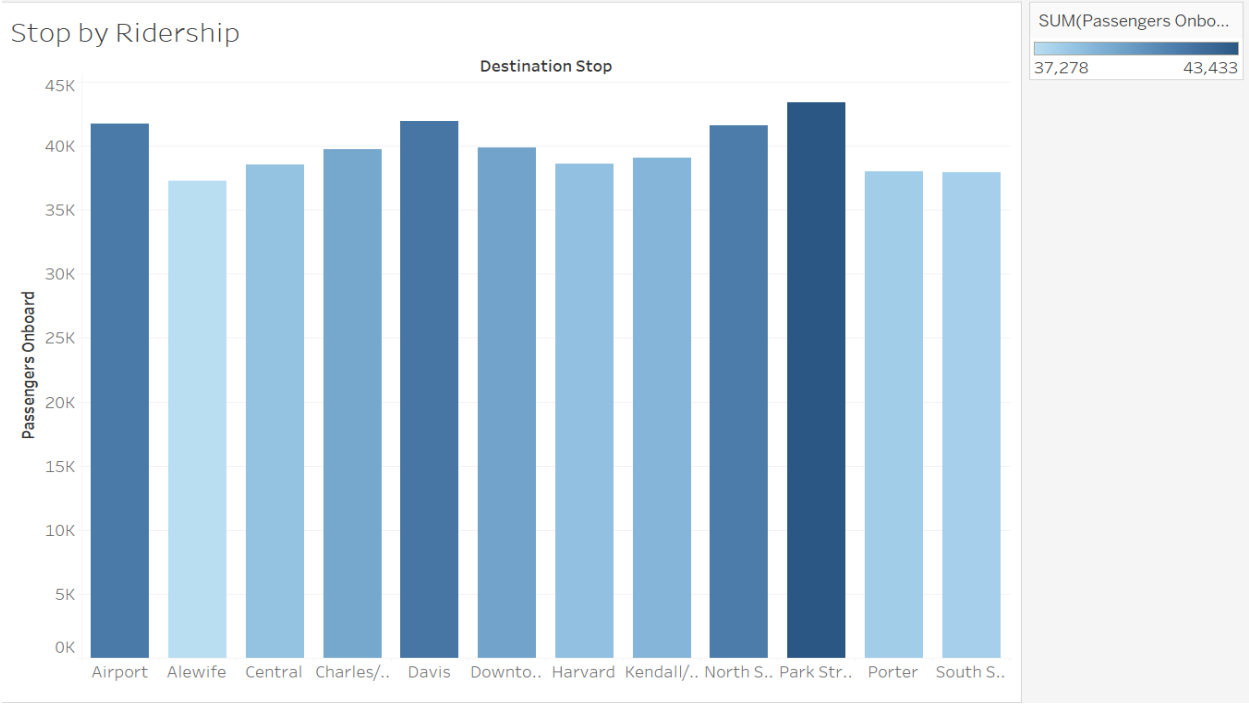

Figure 10

4.c. Resolution: **Root Causes and Costs**: I looked at when and why these delays happen, and what it costs to fix them.

Time Trends: I expected delays to clear up when ridership went down, but the "Delay Trends" graph shows that average delay minutes remain flat from Morning through Late Night, even as passenger volume decreases significantly. Also, Saturday appears to be the worst day for delays, Showing worse efficiency on weekends (Figures 11 & 12).

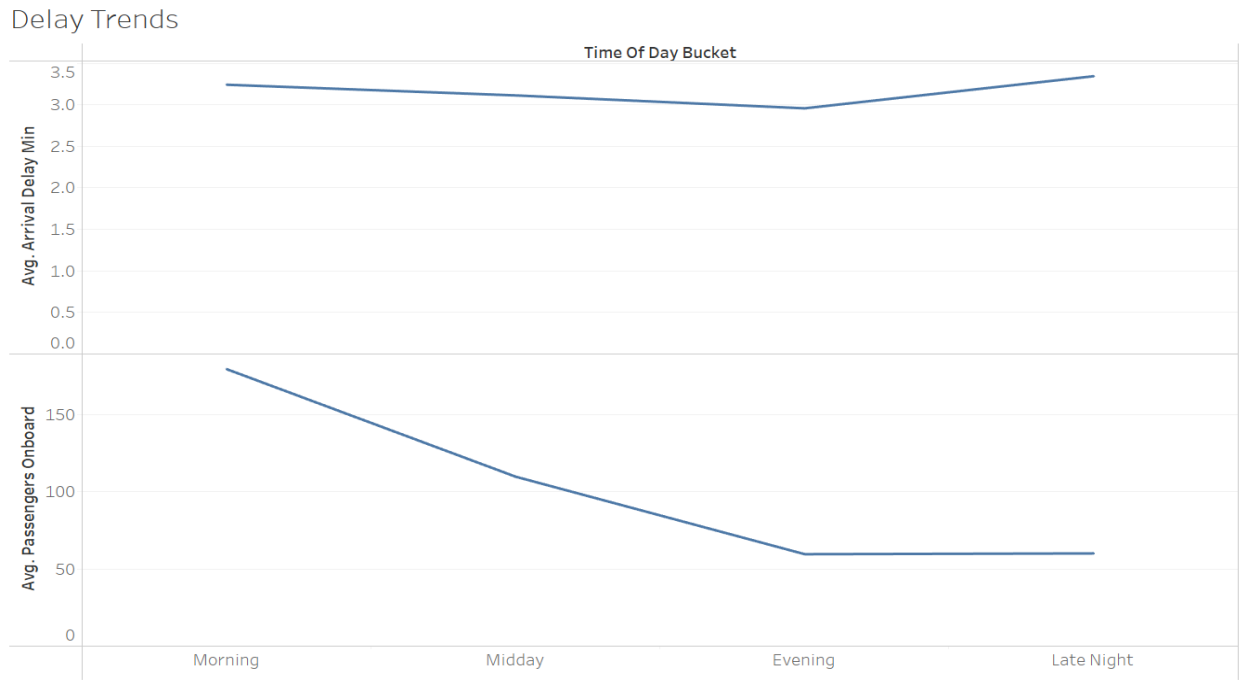

Figure 11

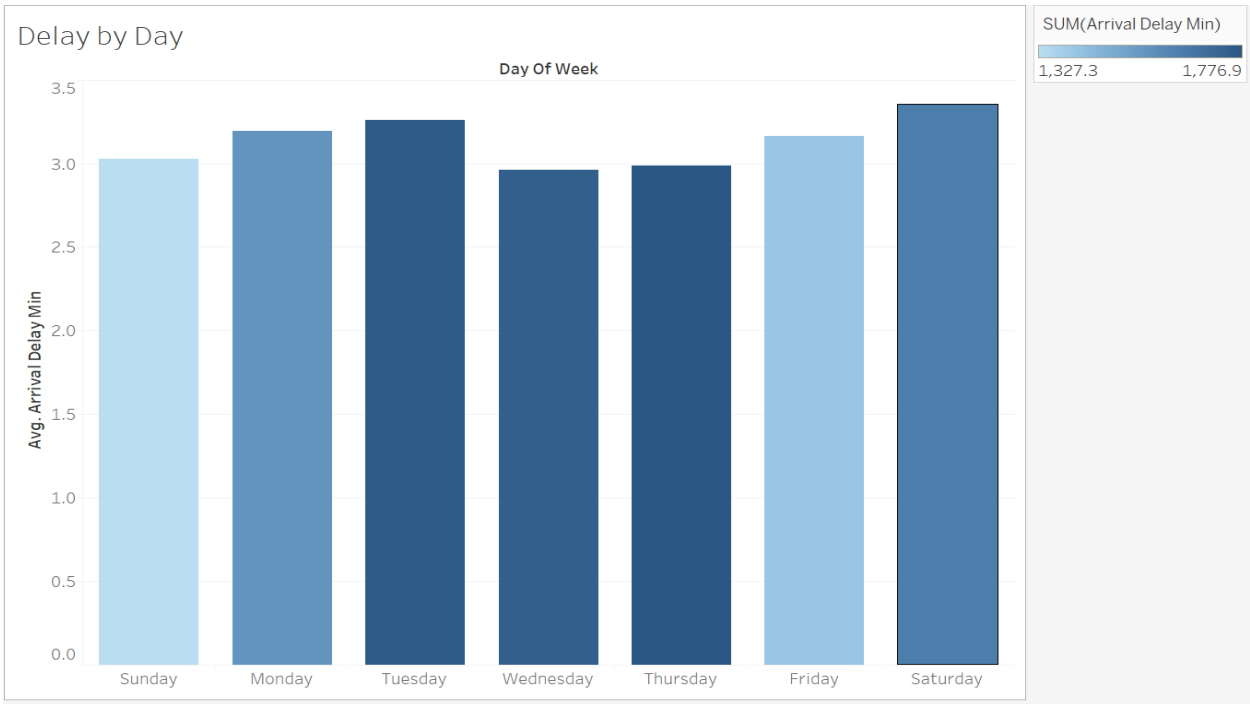

Figure 12

Root causes: The bubble chart shows the percentage of delay minutes by root cause. It reveals that Signal Issues (22.00%), Mechanical Problems (20.13%), and Track Maintenance (15.13%) account for the majority of disruption time (Figure 13).

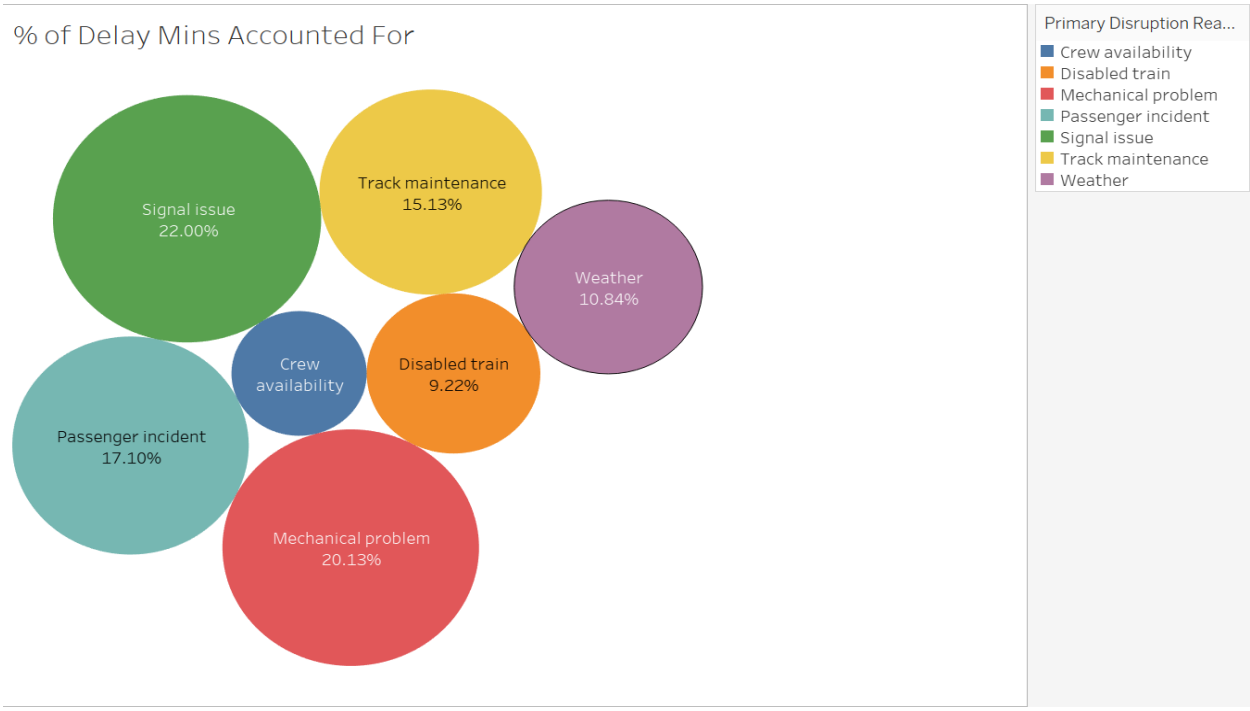

Figure 13

The Cost of Repair: The Python analysis of Work Orders highlights the cost of these failures. With outliers showing crew shifts lasting up to 37 hours and single material costs hitting nearly $3,000, it is clear that when the infrastructure fails, it often requires expensive repairs with heavy labor (Figures 14, 15, & 16).

```
import pandas as pd
import numpy as np
df = pd.read_csv('/content/work_orders.csv')
print(f"Avg hours worked = {df['crew_hours'].mean():,.2f}")
print(f"Max hours worked = {df['crew_hours'].max():,.2f}")
print(f"Total hours worked = {df['crew_hours'].sum():,.2f}")
print(f"Avg cost of job = {df['material_cost_usd'].mean():,.2f}")
print(f"Max cost of job = {df['material_cost_usd'].max():,.2f}")
print(f"Total cost of jobs = {df['material_cost_usd'].sum():,.2f}")
```

```
Avg hours worked = 7.57
Max hours worked = 37.00
Total hours worked = 75,651.40
Avg cost of job = 303.42
Max cost of job = 1,919.27
Total cost of jobs = 3,034,185.66
```

Figure 14

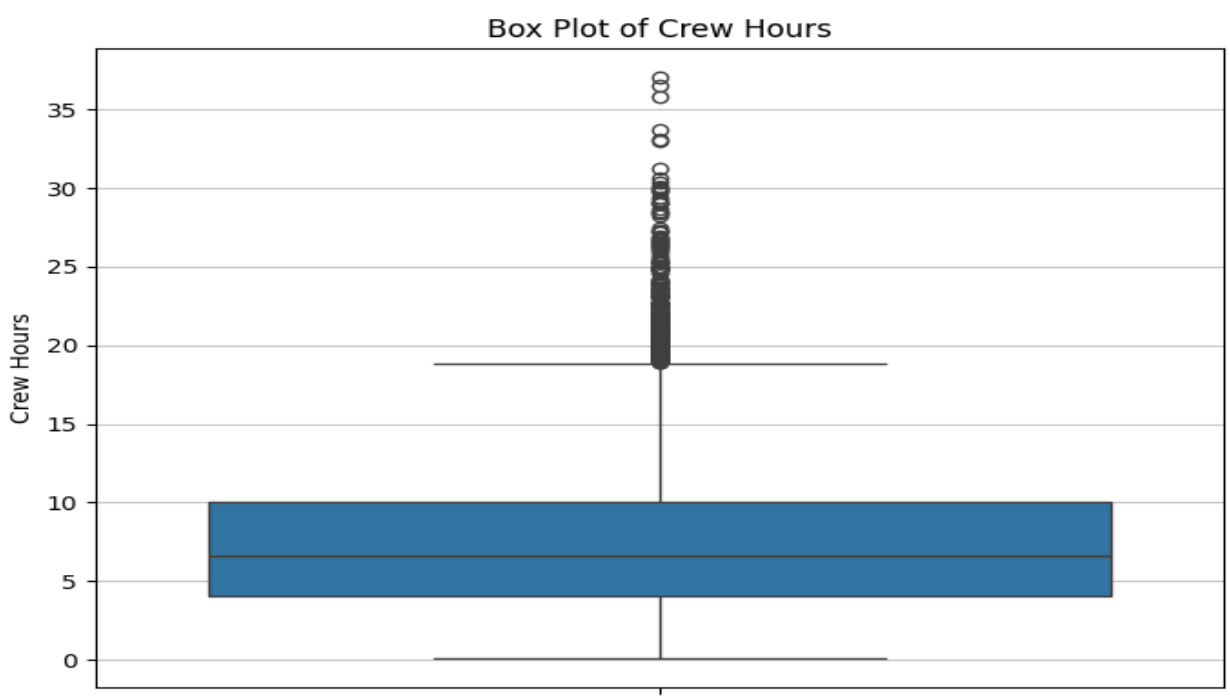

Figure 15

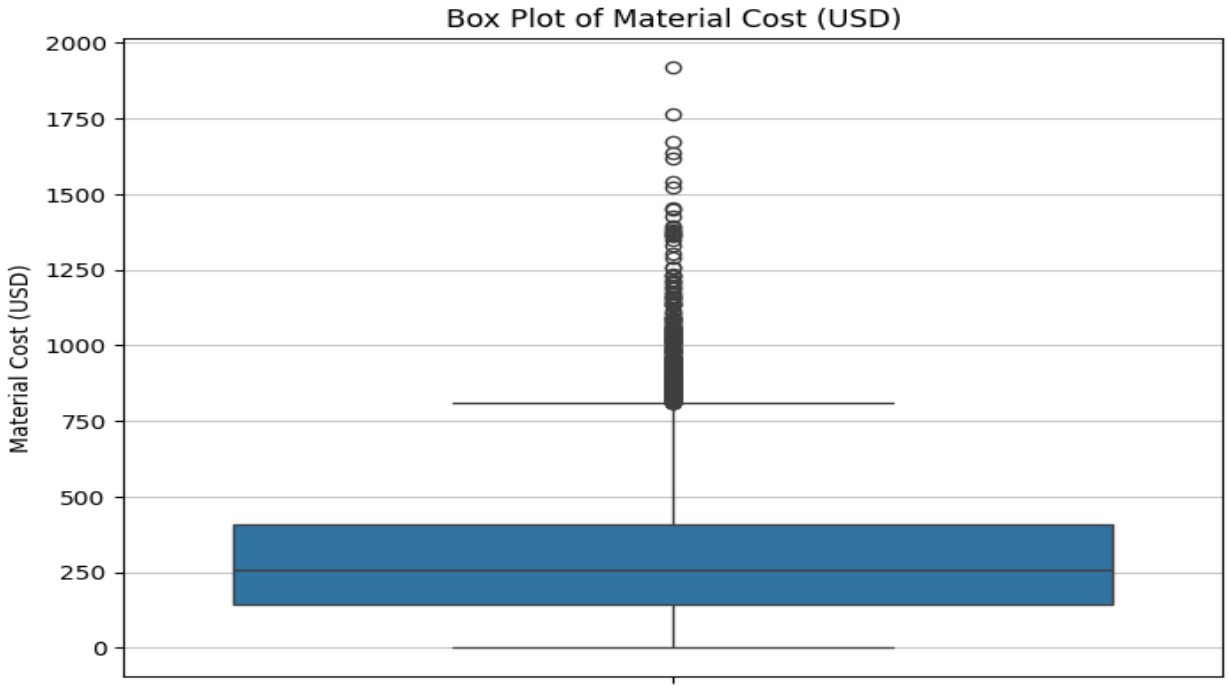

Figure 16

5. **Recommendations** Based on the data story outlined above, I propose the following recommendations to the MBTA:

Modernize Rail Infrastructure: Over 42% of delay minutes are caused by signal issues and mechanical problems, therefore the MBTA should prioritize investments in upgrading signal systems and maintenance. These are internal failures under the MBTA's control, unlike traffic.

More Bus Lanes: The data proves that bus reliability is almost entirely compromised by traffic. To fix the high severity of bus alerts, the MBTA should create more dedicated bus lanes and separate their transit from general congestion around Boston.

Weekend Operations: The spike in delays on Saturdays suggests a lapse in weekend planning by the MBTA. Service and maintenance schedules should be adjusted to ensure weekend reliability matches weekday standards.

Predictive Maintenance: The extreme outliers in the work orders such as 37 hour shifts shows horrible equipment failures that cause long repair sessions. Moving to a predictive maintenance model could catch these issues earlier, turning emergency overtime repairs into routine and cheaper fixes.

6. **Discussion:** The analysis highlights that reliability is lacking in the MBTA. Bus riders are often stuck in traffic. And rail takers are at too high of a risk of getting stuck on trains or waiting for trains due to poor infrastructure.

The finding that delays are consistent throughout the day despite drops in ridership shows that operational inefficiencies come from the MBTA itself and not just a result of "rush hour" crowds. Additionally, the disconnect between riders liking their "most recent trip" but distrusting the "reliability" of the system shows that while the MBTA often works, the fear of a breakdown is a big deal and real, driven by experiences or what people have seen and heard.

7. **Conclusion**: Is the MBTA providing reliable service? The answer is not consistently.

Reliability is significantly compromised by traffic for buses and infrastructure for rail.

Predictability is often incorrect due to the high volume of service alerts.

Satisfaction is weak. Riders are happy when it works, but overall trust is low.

By focusing on the recommendations above, and putting the fees they collect towards improving the system, the MBTA can change from reacting to repairs and be better at predicting, which will increase overall reliability and general perception of the MBTA.

8. **References**

"MBTA Blue Book Open Data Portal." **MBTA-MassDOT Open Data, https://mbta-massdot.opendata.arcgis.com/search?tags=performance. Accessed *[12/12/25]*.**